

Abdul Rahman Shaikh

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AI/ML Engineer | LLM/RAG Systems, Model fine-tuning & Evaluation | Applied ML

SUMMARY

AI/ML Engineer with 5+ years building reliable LLM systems across RAG, fine-tuning, multimodal AI, and inference optimization. Delivered source-grounded AI workflows that reduced document and feedback review cycles from days to minutes, improved retrieval quality by 26%, increased precision@5 to 82%, and boosted LLM throughput up to 4x. Strong in deployment-ready AI systems using Python, Hugging Face, FAISS, Docker, MLflow, Ragas, TruLens, and DeepEval.

SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript, SQL, R
- **ML/AI:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, Keras, OpenCV, CLIP, ViT, SAM
- **LLM/RAG:** RAG, LoRA/QLoRA, RLHF/DPO, RAG, prompt engineering, embeddings, LLM evaluation, vector search
- **Cloud & MLOps:** Docker, FastAPI, AWS, GCP, MLflow, Weights & Biases, DVC, Git, CI/CD, FAISS, REST APIs
- **Evaluation:** precision@k, SHAP/LIME, hallucination/bias testing, Ragas, TruLens, DeepEval, LLM-as-judge
- **Agents & Orchestration:** CrewAI, AutoGen, LangChain, LangGraph, ReAct, tool-calling, MCP

PROFESSIONAL EXPERIENCE

VISUAL TECHNOLOGIES

Plano, TX (Remote)

AI Engineer

2024 – Present.

- Reduced customer-feedback and document-review cycles from multi-day manual analysis to under 5 minutes by architecting RAG pipelines that transform support logs, feedback, and internal docs into source-grounded summaries; deployed as a FastAPI microservice on AWS ECS handling 5K+ daily queries at p95 latency < 900ms.
- Improved retrieval quality from a 61% MRR baseline to 82% precision@5 (+26%) by systematically tuning chunking strategy, embedding model selection (BGE-M3 vs. OpenAI ada-002), metadata filters, cross-encoder reranking, and golden-set evaluation across 500+ query-document pairs.
- Reduced hallucination rate by 38% by implementing citation-grounded generation, answerability checks, and a regression test suite (200+ cases) evaluated via Ragas, TruLens, and LLM-as-judge scoring, ensuring production regressions were caught before every release.
- Achieved 70% preference-evaluation win rate by fine-tuning Llama-3-8B and Mixtral-8x7B with LoRA/QLoRA on 2xA100 GPUs across 50K domain-specific instruction pairs, cutting GPU memory from 48 GB to 19 GB and reducing training time by 45% versus full fine-tuning.
- Increased LLM inference throughput up to 4x by benchmarking and deploying dynamic batching, INT4/INT8 quantization, and speculative decoding, reducing per-token cost by ~55% for resource-constrained production deployments.

NORTHERN ILLINOIS UNIVERSITY

DeKalb, IL

AI/ML Researcher

2020 - 2024

- Architected an LLM-assisted cross-view data exploration system evaluated through a controlled user study (25+ participants) that reduced analyst task-completion time by 35% and improved insight discovery accuracy by 45% versus baseline non-assisted workflows.
- Built multimodal representation-learning pipelines across 1M+ records and 300K+ images using CLIP, transformer encoders, topic modeling (BERTopic), and FAISS vector indexing, supporting large-scale analysis and retrieval for analysis systems.
- Evaluated and benchmarked five candidate models (logistic regression, gradient boosting, random forest, SVM, neural net) against latency, interpretability, and AUC before selecting XGBoost for deployment, achieving 88 AUC at 20ms p95 inference.
- Developed predictive models across 800K+ scholarly records using citation, social media, YouTube, and altmetric signals to forecast patent impact and public engagement, achieving 94.5 F1 on held-out evaluation data.
- Standardized reproducible ML workflows across 3 labs, covering ingestion, feature extraction, modeling, evaluation, and reporting, supporting 10+ peer-reviewed publications; mentored 5+ researchers and led internal workshops adopted as lab-wide best practices.

NORTHERN ILLINOIS UNIVERSITY

DeKalb, IL

Research Assistant

2018 - 2020

- Built applied-ML models for research-impact prediction integrating YouTube (180K+ videos), Twitter, and scholarly metadata, contributing to peer-reviewed publications in Scientometrics and IEEE TVCG.
- Designed end-to-end modeling pipelines (data collection via API/scraping, cleaning, feature engineering, cross-validation, and visualization) that reduced experiment iteration time by ~40% across 4 concurrent research projects.

AMAZON

Hyderabad, India

Data Analyst

2016 - 2017

- Reduced query latency by 25–30% by optimizing SQL and MongoDB pipelines processing 500K+ daily records across large-scale e-commerce and operations datasets.
- Cut recurring analysis turnaround from days to hours by automating ETL validation workflows using Python, Pandas, and NumPy, eliminating ~15 hours of weekly manual reconciliation.
- Improved operational reporting reliability by identifying workflow bottlenecks, data-quality gaps, and performance issues using SQL, Python, Pandas, NumPy, and scikit-learn.

BOTHOOK

Hyderabad, India

Founding Software Engineer

2015 - 2017

- Automated 60% of repetitive FAQ and lead-capture interactions by building NLP-driven chatbot systems deployed across 7+ client websites and messaging channels, handling multiple monthly conversations.
- Developed Node.js backends, REST API integrations, and dialogue-management workflows supporting end-to-end client onboarding from scoping through live deployments.

SELECTED AI SYSTEMS

- **LLMFlow - Scholarly Document Intelligence Platform:** RAG-based document intelligence platform integrating GPT-4, Claude, and FAISS vector indexing, reducing document review time by 60% across 200+ documents.
- **GenHealth - Multimodal Medical Information Extraction Prototype:** Combined GPT-4, BLIP-2/CLIP, structured clinical fields, and FastAPI microservices, improving diagnostic-information extraction F1 by 30% in experimental evaluation.
- **Rufus - Autonomous Web-to-RAG Data Pipeline:** Built a Selenium-based crawler that extracted, structured, embedded, and indexed web content into retrieval-ready knowledge bases reducing knowledge-base creation time by 30%.

PUBLICATIONS

10+ peer-reviewed publications in AI, ML, visual analytics, and data-driven decision support, including **IEEE TVCG**, **IEEE VIS (Best Paper Honorable Mention)**, **JCDL**, **Scientometrics**, and **Big Data**. Full list available via [Google Scholar](#) and [portfolio website](#).

EDUCATION

NORTHERN ILLINOIS UNIVERSITY

DeKalb, IL

Ph.D. in Computer Science

2020 - 2026 (Expected)

Dissertation: LLM-Driven Cross-View Data Exploration | GPA: 3.9

Teaching: TA for ML, Algorithms, & Databases - led recitations, designed assignments, and graded for 40 students across 8 semesters

M.S. in Computer Science

2018 - 2020

Thesis: Modeling the Broader Impact of Science and Health Using Social Media | GPA: 3.9